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Write legibly showing all the steps VERY clearly. (Total points = 100).

Let $\Omega \neq \emptyset$ and $\mathcal{P}(\Omega)$ is the collection of all subsets of Ω , i.e the power set. Let $\mathcal{G}, \mathcal{C}, \mathcal{F} \subset \mathcal{P}(\Omega)$ be a classes of subsets of Ω , and λ, μ be set functions. ϕ = empty set.

- [1] (a) Define when we say $\mathcal{C}$ is a (i) semi-algebra, (ii) algebra, (iii) σ -algebra.
(5 $\times$ 3 = **15** points)

(i) $\mathcal{C}$ is called a semialgebra if $A, B \in \mathcal{C}$, then $A \cap B \in \mathcal{C}$ and for any $A \in \mathcal{C}$, there exist sets $B_1, B_2, \dots, B_k \in \mathcal{C}$; for some $1 \leq k < \infty$; such that $B_i \cap B_j = \phi$, for $i \neq j$, and $A^c = \cup_{i=1}^k B_i$.

(ii) $\mathcal{C}$ is called a algebra if $\Omega \in \mathcal{C}$, for any $A \in \mathcal{C}$, $A^c \in \mathcal{C}$ and $A, B \in \mathcal{C}$ implies $A \cup B \in \mathcal{C}$

(iii) $\mathcal{C}$ is called a σ -algebra if $\Omega \in \mathcal{C}$, for any $A \in \mathcal{C}$, $A^c \in \mathcal{C}$ and $A_1, A_2, \dots \in \mathcal{C}$ implies $\cup A_i \in \mathcal{C}$

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- (b) Suppose $\mathcal{G} \neq \emptyset$ is a semi-algebra, and also closed under finite *disjoint* unions, i.e. if $A \cap B = \phi$ and $A, B \in \mathcal{G}$, then $A \cup B \in \mathcal{G}$. Prove $\mathcal{G}$ is an algebra. (15 points)

Since $\mathcal{G} \neq \emptyset$, take $A \in \mathcal{G}$. Then there exist sets $B_1, B_2, \dots, B_k \in \mathcal{C}$; for some $1 \leq k < \infty$; such that $B_i \cap B_j = \phi$, for $i \neq j$, and $A^c = \cup_{i=1}^k B_i$. Since $\mathcal{G}$ is closed under finite *disjoint* unions, $\Omega = A \cup (\cup_{i=1}^k B_i) \in \mathcal{G}$. Also for any $A \in \mathcal{G}$, we know $A^c = \cup_{i=1}^k B_i$ where B_i s are disjoint, and hence $A^c \in \mathcal{G}$. Since it is closed under complements and closed under intersections (semi-algebra), using DeMorgan's law, it is closed under finite unions as well.

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- (c) Suppose $\mathcal{F} \neq \emptyset$ is an semi-algebra and is closed under finite disjoint unions. In addition, it is also a *monotone class*: i.e.
if $\{A_i\} \in \mathcal{F}$, $A_i \uparrow$, then $\cup A_i \in \mathcal{F}$ and if $\{B_i\} \in \mathcal{F}$, $B_i \downarrow$, then $\cap B_i \in \mathcal{F}$.
Prove $\mathcal{F}$ is an σ -algebra.

(10 points)

As in part (b), we know $\mathcal{F}$ is an algebra. In addition, because of increasing union property, it is a sigma-algebra, by the theorem in the book. Providing proof of the theorem is optional

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- (d) Suppose λ be a set function on the measurable space $(\Omega, \mathcal{F})$, where $\mathcal{F}$ is a σ -algebra. Let λ satisfy $\lambda : \mathcal{F} \rightarrow [0, \infty]$; $\lambda(\emptyset) = 0$ and it is *finitely additive* and *countably sub-additive* : i.e. if $A \cap B = \phi$ and $A, B \in \mathcal{F}$, then $\lambda(A \cup B) = \lambda(A) + \lambda(B)$; and if $\{A_i\} \subset \mathcal{F}$ then $\lambda(\cup A_i) \leq \sum_i \lambda(A_i)$. Prove that λ is a measure on $(\Omega, \mathcal{F})$

(15 points)

We only need to verify the countable additivity property. Take $\{A_i\}$ a sequence of disjoint sets in $\mathcal{F}$. Then we know from sub-additivity (disjoint-ness is not used):

$$\lambda(\cup_{i=1}^{\infty} A_i) \leq \sum_i \lambda(A_i) \quad (*)$$

Also, note that the set function is monotone, i.e if $A \subset B$ in $\mathcal{F}$, then $\lambda(B) = \lambda(A) + \lambda(B \setminus A) \geq \lambda(A)$. Hence, for each $N \geq 1$,

$$\lambda(\cup_{i=1}^{\infty} A_i) \geq \lambda(\cup_{i=1}^N A_i) = \sum_i^N \lambda(A_i),$$

by finite additivity. Since it holds for each $N \geq 1$, taking limit, we get

$$\lambda(\cup_{i=1}^{\infty} A_i) \geq \sum_i^{\infty} \lambda(A_i)$$

hence by (*) above, the proof is complete

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[2] Let $F(x)$ be the cdf of standard normal (Gaussian) distribution, i.e.

$$F(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du, \quad x \in \mathbb{R}$$

We know that the integrand is symmetric around 0 and the integral does **not** allow a closed-form. Instead, the cdf is numerically computed for intervals and presented in the standard normal table (the so called z-table) part of which is given below.

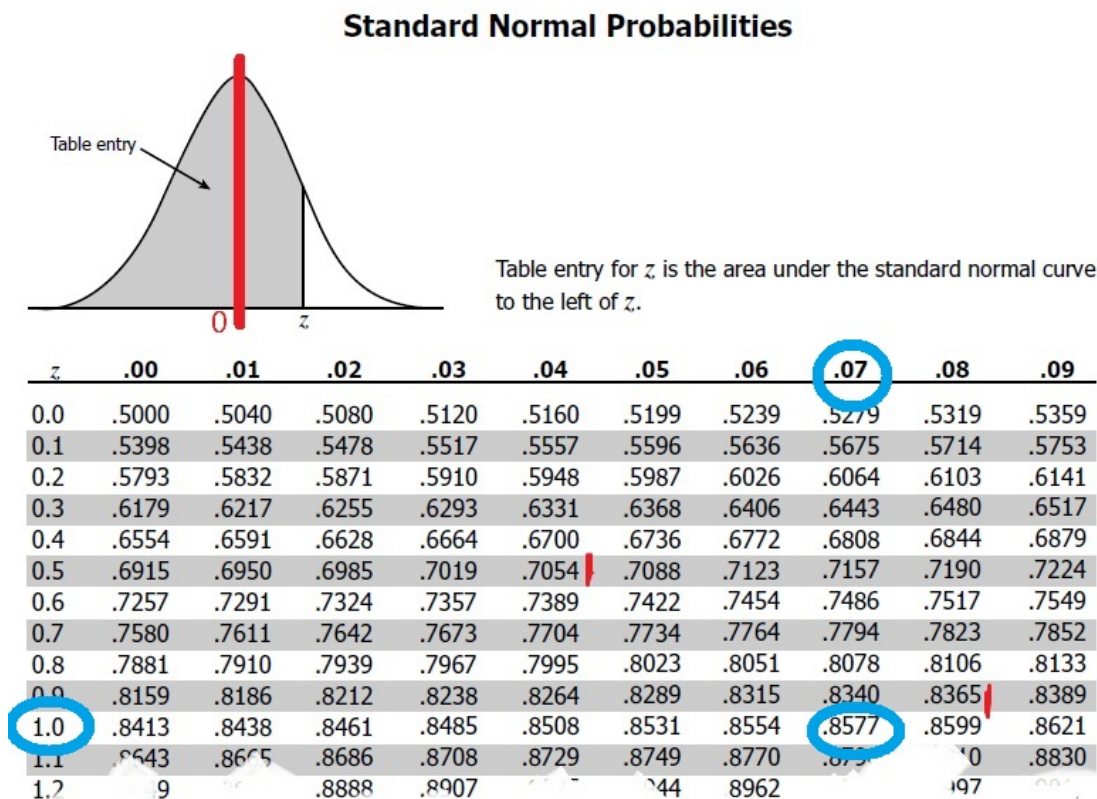


Figure 1: Part of Z-table. This shows, for example, $F(1.07) = 0.8577$.

- (a) Using the steps in Caratheodory Extension Theorem(s), show by starting from defining a measure μ_F on a suitable semi-algebra, how one can extend it to the Lebesgue-Stieltjes measure μ_F^* on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

[Write the steps of the theorems carefully, stating the results used in each step]

(15 points)

(a) Let $\mathcal{C} = \{(a, b] : -\infty \leq a < b < \infty\} \cup \{(a, \infty), a \in \mathbb{R}\}$. Define

$$\mu_F(a, b] = F(b+) - F(a+) = F(b) - F(a); \quad \mu_F(a, \infty] = F(\infty-) - F(a+) = 1 - F(a)$$

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Then this defines a measure on semi-algebra $\mathcal{C}$. μ_F^* restricted to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an extension of μ_F . By Caratheodory Extension Theorem, there exists an extension of μ_F , called μ_F^* on $(\Omega = \mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$, where $\mathcal{M}_{\mu_F^*}$ = set of all μ_F^* -measurable sets on $\Omega = \{A \in \mathcal{P}(\Omega) : \mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^C), \forall E \in \Omega\}$. $\mathcal{M}_{\mu_F^*}$ is a σ -algebra containing $\mathcal{C} \Rightarrow \mathcal{B}(\mathbb{R}) \subset \sigma\langle \mathcal{C} \rangle \subset \mathcal{M}_{\mu_F^*}$. Hence, the restriction of μ_F^* to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is the extension.

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(b) Why is such a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ unique?

(10 points)

This is unique since $\mu_F^*(\mathbb{R}) = 1$ (finite measure), Hence by the uniqueness Theorem, any other measure that takes same value on $\mathcal{C}$ will take same values on $\sigma\langle\mathcal{C}\rangle = \mathcal{B}(\mathbb{R})$. This is the Normal (0,1) or standard Gaussian measure.

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- (c) For this extended measure μ_F^* (or μ_F), find $\mu_F^*(\{x \in \mathbb{R} : |x| \geq 0.5\})$ (10 points)

Let $A = (-\infty, 0.5] \cup [0.5, \infty)$. So, by the symmetry of the density, i.e. by the symmetry of the integrand in the cdf, $\mu_F^*(A) = 2\mu_F^*([0.5, \infty)) = 2(1 - \mu_F^*((-\infty, 0.5])) = 2(1 - \mu_F^*((-\infty, 5])) = 2(1 - 0.6955) = 0.6160$

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- (d) Let m be the Lebesgue measure (i.e. Lebesgue Steiljes measure with $F(x) = x$). We know already that m is translation and reflection invariant. In addition, it satisfies the scaling property:

$$m(cA) = |c|m(A), \quad c \in \mathbb{R}, A \in \mathcal{B}(\mathbb{R}).$$

Prove this scaling property.

(10 points)

Fix $c \in \mathbb{R}$ and define the set functions $\mu_c^1(A) = |c|m(A)$ and $\mu_c^2(A) = m(cA)$ for all $A \in \mathcal{B}(\mathbb{R})$. It is easy to check that μ_c^i is a valid measure, and on $\mathcal{C}$ (as in part (a)), $\mu_c^1(A) = \mu_c^2(A)$ for all $A \in \mathcal{C}$. To check this, note that $m(c(a, \infty)) = \infty = |c|m((a, \infty))$ and $m(c(a, b]) = m((ca, cb]) = |c|(b - a) = |c|m((a, b])$. By uniqueness theorems and since $\mathcal{B}(\mathbb{R}) = \sigma\langle\mathcal{C}\rangle$, $|c|m(A) \equiv \mu_c^1(A) = \mu_c^2(A) = m(cA)$ for all $A \in \mathcal{B}(\mathbb{R})$. This completes the proof.

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Additional page (attach more sheets if needed. Number all additional sheets.)